Patrick Thangarajah

FINITE ELEMENT ANALYSIS | COMPUTATIONAL BIOMECHANICS | LS-DYNA | MODEL VALIDATION

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Summary

Recent MASc graduate in mechanical engineering with experience in vehicle safety, impact biomechanics, finite element modelling, physical test data analysis, literature review, and technical writing. Graduate research focused on validating a THOR-5F FE model and evaluating how seat-belt anchor locations affected rear-seat occupant response in frontal impacts.

Skills

Occupant Safety and Crash Analysis	LS-DYNA, nonlinear FEA, explicit dynamics, crash simulation, injury biomechanics
Model Development and Validation	FE model validation, physical test correlation, ISO 18571, material models, contact, boundary conditions
Simulation and Data Analysis	LS-PrePost, MATLAB, JMP, design of experiments (DOE), Python, Microsoft Excel
CAD and Engineering	SolidWorks, engineering drawings, technical reports, technical presentations

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in occupant safety and biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Work/Research Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Validated a THOR-5F ATD finite element model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Investigated material response, contact, boundary conditions, occupant positioning, and restraint fit to understand simulation-to-test discrepancies
- Designed and analyzed a 104-simulation DOE-based FE parametric study evaluating the effect of seat belt restraint anchor locations on rear seat occupant response, identifying a rearward and inboard position that reduced peak chest deflection by 16% below the study average
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*
- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Created standardized work instructions and SOPs verified against engineering drawings and customer specifications
- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time

Projects

Effect of Seat Foam Properties on Injury Response in Rear Impact (ME 725 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Jan 2024 - Apr 2024*
- Verified 5 polyurethane foam material models in LS-DYNA against input stress-strain response using single-element simulations, then implemented them in a rear-impact sled model with a Hybrid III ATD to calculate HIC15 and Nij injury metrics
- Determined that all 5 foams met the regulatory head injury limit but exceeded the regulatory neck injury limit at high speed, and that the softest foam reduced head injury risk by roughly 18–20% relative to the stiffest

Simulation of Cervical Fractures in American Football (ME 720 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Dec 2023*
- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.